

Automated Pothole Detection and Severity Grading Using Transfer Learning

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Abstract—Potholes represent a critical global road infrastructure challenge, posing significant risks to public safety and often causing major traffic incidents. Consequently, road users actively seek to avoid them, and municipalities require efficient ways to manage repairs. By leveraging artificial intelligence, it is possible to automate the assessment of pothole severity, distinguishing between negligible flaws and high-risk hazards. This study compares two distinct deep learning approaches for classifying pothole severity: the single-stage YOLOv8 Segmentation model and a Two-stage Heuristic Convolutional Neural Network (CNN). The objective is to evaluate and determine which architecture provides superior accuracy and efficiency for real-world severity classification

Keywords—potholes, YOLOv8, severity, deep learning, segmentation, Two stage Heuristic CNN,

I. INTRODUCTION

Potholes present a pervasive global infrastructure challenge, with municipal repairs often restricted by resource constraints to a reactive basis. Consequently, the burden of traffic safety shifts to road users who must actively navigate these hazards. To mitigate the risk of vehicle damage and road incidents, automated classification of pothole severity is essential. This study evaluates deep learning architectures to automate road defect categorization, specifically comparing a single-stage YOLOv8 Segmentation model against a Two-stage Heuristic Convolutional Neural Network (CNN). Ultimately, this research aims to identify the most effective architecture for real-time driver assistance and automated infrastructure monitoring.

II. LITERATURE REVIEW

Traditional road monitoring relies on manual inspections or vibration sensors [1], which are either slow or prone to false alarms. Today, vision-based deep learning is the standard approach. Transfer learning, in particular, allows pre-trained models to detect potholes quickly and accurately with less computational cost [2]. However, while detecting a pothole is straightforward, grading its physical severity from a standard 2D camera remains a major challenge. Because 2D images lack depth information, researchers often rely on expensive 3D sensors like LiDAR to measure volume [3]. Developing a transfer learning model that can grade severity using only monocular vision provides a much more affordable and scalable solution.

To successfully train a model for severity grading, the dataset needs a standardized ground truth. This project adopts the ASTM D6433-16 standard for Pavement Condition Index (PCI) surveys [4]. Under this standard, pothole severity is strictly categorized into Low, Medium, or High based on an intersecting matrix of maximum depth and average diameter. For example, a pothole with a depth between 13 mm and 25 mm and a diameter between 100 mm and 200 mm is classified as Low severity. Conversely, a pothole exceeding 50 mm in depth and 450 mm in diameter is a High severity hazard [4]. Using this engineering standard ensures the dataset is labeled accurately, allowing the model to learn physical severity metrics rather than subjective visual estimates.

III. METHODOLOGY

A. Ground Truth and Dataset Establishment

The proposed dataset consists of dashcam-perspective imagery specifically curated to represent real-world driving conditions, including varying lighting, shadows, and road textures. The ground truth for each pothole was manually annotated by establishing bounding boxes and assigning severity classifications based strictly on spatial dimensions and visual footprint. This ground truth serves as the baseline benchmark to evaluate the classification accuracy and F1-scores of the three proposed automated models.

B. Severity Estimation Heuristics

Across all tested methodologies, the final severity classification relies on a shared heuristic algorithm that maps 2D pixel data to physical threat levels. Once a region of interest (ROI) is established, the system extracts a Size Proxy Score S_{size} to determine the hazard level.

Perspective-Aware Scaling: To account for the monocular domain gap, where objects appear smaller as they approach the vanishing point, the raw pixel area of the detected hazard is dynamically scaled based on its vertical spatial coordinate (y-axis) within the frame. The proximity of the bounding box or mask to the bottom of the frame (the camera's immediate foreground) dictates the scaling factor applied to the raw area. This normalized S_{size} is then evaluated through a discrete classification threshold matrix to assign the final severity of Low, Medium, or High.

C. Benchmarked Architectures

To determine the optimal balance between computational efficiency and classification accuracy, two distinct extraction methods were evaluated against the dataset ground truth.

1) Method 1: Two-Stage Heuristic Convolutional Neural Network (CNN)

This two-stage approach optimizes processing speed. A CNN acts as a detector to isolate potholes. Once localized, severity is calculated using a heuristic formula based on the bounding box dimensions relative to their position in the camera frame, bypassing the need for complex segmentation processing.

2) Method 2: YOLOv8 Instance Segmentation

This method employs YOLOv8 to generate a precise pixel-level mask for each pothole. Because the mask inherently excludes background asphalt, the severity classification is derived directly from the exact pixel footprint of the mask. This provides the most accurate spatial measurement of the three methods, as it eliminates noise found in bounding-box approximations.

IV. Results And Discussion

The experimental analysis compared the performance of two architectural approaches: the Two-Stage Heuristic CNN and the Instance Segmentation model. The comparative metrics are summarized in Table I.

Table I: Comparative Performance Metrics

Metric	Instance Segmentation	2-stage Heuristic CNN
Precision	0.308	0.197
Recall	0.105	0.110
mAP@0.5	0.0726	0.0559
mAP@0.5:0.95	0.0726	0.0159

The results demonstrate that while both models encounter challenges with the complex geometry of the proposed dataset, the Instance Segmentation architecture provides a measurable advantage in spatial reliability.

- 1. Precision and Localization:** The Instance Segmentation model achieved a significantly higher precision (0.308) compared to the Heuristic CNN (0.197). This confirms that pixel-level mask generation is more effective at filtering out non-hazardous road features than standard bounding-box approximations. The most significant finding lies in the mAP@0.5:0.95 metric, where Instance Segmentation (0.0726) outperformed the Heuristic CNN (0.0159) by a factor of nearly five. This indicates that the segmentation masks provide a much higher degree of localization accuracy, ensuring that hazard boundaries are more

reliable for downstream severity assessment.

2. **Severity Classification:** The confusion matrix (Figure 1) illustrates the current classification behavior. The model exhibits a bias toward the "minor" severity category, indicating that the heuristic logic is currently calibrated to be conservative. This suggests that the current classification threshold may be overly sensitive to minor surface variations, requiring further optimization in the heuristic grading logic.

The lower metrics across both models are indicative of the extreme difficulty of the dashcam-perspective dataset, which provides a rigorous benchmark for future hazard classification systems. The performance parity in Recall suggests a "precision-recall trade-off", while the CNN maintains a slightly higher recall (0.110), the Instance Segmentation model's superior precision and higher mAP scores demonstrate that it is the more stable architecture for autonomous road maintenance. These findings establish that instance-based feature extraction is the optimal path for high-accuracy road hazard assessment, providing a robust foundation for future data-centric refinements.

V. CONCLUSION

As a conclusion, even though the original objectives couldn't be met, the author managed to complete a working video segmentation pipeline with severity overlay and also completed the pothole severity classification dataset with a total 296 images with labels 0,1, and 2 which directly represent minor, moderate and severe pothole respectively.

As shown in the final results, the model was not able to validate the drone-style imagery data which affects. This is also added with the fact that the severity classification pipeline is rule-based and not learned within the model.

In the future, several improvements can be done, firstly being to finetune the model on a more generalised dataset to consider different kinds of road condition, different angle of camera and also the distance of the camera from the road. This aims to improve the model into becoming more robust and perform with higher accuracy. Secondly, the

severity classifier could be replaced with a neural network classifier that takes the masked region as input. This would also improve the model as it will focus more on the depth and the textures. Lastly, to train the model with more epochs and variability to further generalise the model.

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